Applying Terrestrial Geo-Material Science Methodology to Lunar ISRU Construction

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ABSTRACT

Building on research underway for military and expeditionary construction, this research aims to establish methods for the design and testing of the materials mixtures derived from the lunar surface with minimal or no additives that need to be transported from the Earth. This paper will review opportunities and challenges of demonstrated and experimental construction methods such as concreting and sintering as well as discuss the need to evaluate and develop these methods using regolith simulants more representative of the lunar highlands where the NASA Artemis mission proposes to land in both its mineralogical and agglutinated properties. Advanced materials research based on accurate chemical and physical characteristics of extraterrestrial materials is among the first steps towards establishing a methodology that will form a proposed future ASCE Manual of Practice for evaluating and implementing best practices for in-situ lunar construction.

INTRODUCTION

As the nation and the world prepare for the first lunar return in nearly 50 years, a pressing need exists to develop and test construction methods that use in-situ (locally available) materials for critical infrastructure. Building any exploratory base camp requires site preparation for horizontal (e.g., landing pads, roads) and vertical structures (e.g., berms, blast and radiation shielding).On Earth, building such infrastructure requires performing an in situ geotechnical site assessment, which is a luxury not afforded on the Moon. This highlights a knowledge gap in how to measure and predict the lunar geotechnical properties necessary for in-situ engineering and construction and prepare the lunar surface for permanent occupation.

Generating building products from materials available in situ (locally available) is required for the next phase of exploration, development, and occupation of extraterrestrial surfaces. The scale and complexity of the operations required for resource extraction and human inhabitation preclude the exclusive use of imported resources from Earth. Rather, such operations require use of indigenously developed materials through a practice commonly known as in-situ resource

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utilization (ISRU). ISRU will become increasingly important as space exploration evolves from resource discovery to resource development, and ultimately to sustained human occupation of planetary surfaces. Future astronauts need the ability to collect space-based resources and transform them into breathable air, water for drinking, hygiene, and plant growth; rocket propellants, and building materials.

By comparison, for austere and expeditionary applications, the US military uses stabilized soil and aggregate systems (SSAS) which offer advantages of expediency of construction and reduced materials and construction costs compared to conventional infrastructure materials. SSAS have proven to be able to provide the engineering properties required for use as embankments, dams, roadway and airfield pavement components, rapid repair materials, revetments, etc. Analogous to ISRU construction, SSAS counts on locally available mineral aggregates and/or soils as the major component of the SSAS product. Therefore, it is imperative to be able to predict the products that will result from the interactions of the chemical stabilizers and the mineral constituents, and work has begun to design a methodology for SSAS using both inorganic chemical additives (often referred to as geopolymers) as well as organic polymers to provide targeted engineering properties. The methodology incorporates thermodynamic based methods and multiscale experimentation and modeling to predict and quantify the products that provide the strength and durability of the SSAS. Of equal importance, this method will assess the potential of any deleterious product development that would hamper the success of the product.

To put ISRU and in situ construction into practice, a comparable need exists to determine material processing methods using accurate extraterrestrial surface materials. How do lunar regolith geochemical properties affect the stability of hybrid materials/composites? How might these properties vary when using regolith simulants for the lunar highlands— where Artemis missions are targeted to land— vs those from the mare regions? How does the more highly agglutinated content of these highlands affect these properties? How might basalt glass fiber— which can be readily derived from lunar regolith— perform as a building material? Are different methods or materials needed for different applications, such as landing pad construction or radiation shielding? This paper begins to discuss these topics as well as some initial steps to undertake in order to begin providing answers.

Background: Lunar surface environment— Terrain and material

The general landscape of the Moon is controlled by high-velocity impacts and volcanism and is composed of hills, impact craters, volcanoes, and lava flows. The entire surface of the Moon is covered in a layer of material known as regolith. Lunar regolith is composed of dust and rock fragments and is created by the constant bombardment of the lunar crust by meteoroids, solar UV flux, solar wind, and radiation over billions of years (McKay et al., 1991).

There are two major terrain classes on the Moon: the mare and the highlands. The mare takes up ~15% of the lunar surface and is composed of basaltic lava flows. It is relatively young, resulting in a less cratered and "flat" surface, but tends to have a higher concentration of boulders because the regolith layer is thinner, allowing meteorite impacts to excavate bedrock from beneath. The second major terrain class—the highlands – is composed of anorthosite which is the original lunar crust. The highlands are much older than the mare and have become more heavily cratered over the lifetime of the Moon. Their greater age means that the regolith layer is much thicker which prevents impacts from excavating bedrock from below. Consequently, the highlands have fewer exposed boulders than the mare (Vaniman et al., 1991). Impact crater walls

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and the transition zones between the mare and highlands terrain contribute to most of the relief and the greatest slopes on the lunar surface, but steep slopes are generally uncommon.

Regolith is described as broadly-graded silty sands or sandy silts that are very fine-grained. Of the bulk lunar regolith, ~95% by mass is <1.37 mm and ~50% by mass is <50 μm with a mass average particled size of 72 μm and a bulk density range from 1.30 g/cm^3 to 1.92 g/cm^3 (Carrier, 2005). The lunar regolith has two major particle size fractions within it: soil and dust. The lunar soil, which is the <1 mm particle size fraction takes up ~70% by mass of the regolith by weight. The lunar dust, the <20 μm size fraction, comprises ~20% by mass. The finest particles within the lunar regolith are less than 1 μm in size, and account for ~1% by mass. (Taylor, 2007).

The chemical composition of the lunar regolith varies significantly from a titanium-rich basaltic composition in the mare to a plagioclase feldspar-rich anorthosite in the highlands (Fig. 1a). Water and volatile elements are present in the form of ice in the regolith within the permanently shadowed regions (PSRs) at the lunar poles. The shape of the lunar regolith particles varies from spherical to extremely angular. Individual particles can be elongated and very irregular, which leads to higher compaction of the grains at depth (Carrier et al., 1991). The first ~10cm to 20 cm of the regolith is loose and uncompacted and behaves similarly to dense beach sand. Just below that loose upper layer is a dense, highly compacted layer of regolith. This is due to the vibrations experienced from the constant bombardment of the surface, causing the irregularly shaped grains to interlock (Fig 1b). The interlocking of the grains increases the cohesion of the regolith and allowed the Apollo astronauts to dig trenches into the regolith with smooth, nearly vertical walls to a depth of ~3 m (Carrier, 2005).

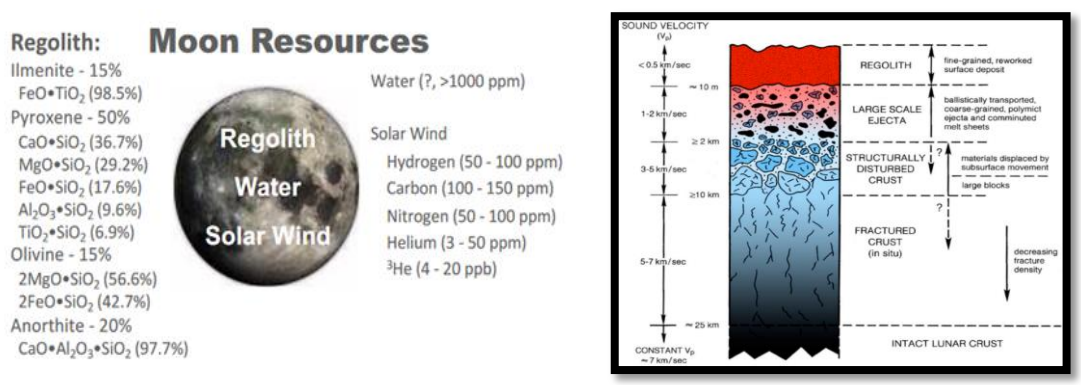

Figure 1a and 1b: Moon Resources (regolith makeup) and Lunar Stratigraphy (Used with permission: Hortz F. et al, 1991)

Agglutinates

The lunar regolith is composed of five general particle types: mineral fragments, crystalline rock fragments, breccia fragments, various glasses, and unique particles called agglutinates. Lunar agglutinates are fused particles of impact glass, rock, and mineral fragments formed when micrometeorites impact into the regolith, melting the regolith on contact to create impact glass. Chemically, the particles typically reflect the local rock type in which they form. However, agglutinates contain relatively high abundances of solar wind gases such as H and He and always contain small droplets of pure Fe metal, also called nanophase iron (NP-Fe), which gives the regolith its magnetic properties. These loosely bound glassy particles are typically <1 mm in

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size, irregularly shaped, and break down easily under load. Agglutinate content increases with regolith maturity and exposure age, accounting for anywhere between 5% and 65% by volume (25% to 30 vol% on average) of the bulk regolith (McKay et al., 1991). Therefore, older surfaces such as the highlands terrain will have a higher agglutinate content. Agglutinates introduce unique chemical and physical properties of the regolith that likely affect their performance in materials mixtures for in situ construction.

Lunar surface construction— Methods and considerations

Most lunar construction materials made by ISRU will require on-site consolidation of granular media into monolithic, load-bearing structural elements. The two primary methods to consolidate granular media are concreting and sintering. Simply put, concreting involves binding grains or aggregate within a chemically activated matrix. Sintering is the fusing of fine grains together by the application of heat to reduce the surface energy. Either or both of these methods can be used on the Moon, but the lunar environment poses challenges that must be addressed to make them robust.

Similarly, additive construction is based on either concreting— by extrusion through a pump— or sintering— by selectively heating a powder bed. It requires meticulous control over rheological vs. hardening properties (viscosity vs. cure time) or heat transfer/particle size distribution/ powder flow (sintering). Additive construction methods, therefore, would be informed by this methodology to characterize regolith-based feedstock mixtures with a more heightened degree of process control.

Concreting Methods

Concrete is a probable candidate for lunar construction because it can potentially produce large volumes of material with complex shapes using relatively low-tech methods and supplies. Moreover, the basalt and anorthosite rocks on the Moon are ideal candidates for the coarse aggregate component that provides much of the mechanical integrity of concrete (Fig 2).

Figure 2. Comparison of Basalt, Anorthosite, and OPC (Used with permission: Nyung. D et al. 2014)

On Earth, concrete is made by binding sand and coarse aggregate within a calcium-rich hydraulic cement that hardens when mixed with water in approximately equal mass ratios. This

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immediately highlights two challenges for lunar applications: (1) the relative scarcity of calcium in lunar regolith and (2) the scarcity of water on the Moon. The scarcity of calcium-based minerals can be overcome by using pozzolanic, geo-polymeric cements instead. Already being used commonly in terrestrial environments, geopolymers are produced from aluminosilicate precursors— much like those on the Moon (Pilehvar et al, 2009)— by breaking down their existing molecular structure in a chemical activator and allowing them to re-condense in a gel-like polymeric structure that is remarkably strong and chemically stable.

The second challenge, scarcity of water, is a more serious obstacle to concreting. All terrestrial concrete, whether based on traditional calcium-rich components like Portland cement or on geopolymers, require large volumes of liquid water, at least 20% of the total mass, to initiate and sustain the necessary chemical reactions. Water locked in regolith as ice in the polar regions is likely needed as propellant and to sustain human life. Therefore, concreting methods can be viable only if a way is found to manufacture liquid water on the Moon, or if some alternative liquid binding agent available on the Moon can be identified.

One way to produce liquid water is to extract oxygen from lunar regolith and combine it with hydrogen transported from Earth. Oxygen extraction from lunar regolith has been demonstrated by carbothermal reduction (Gustafson, R., et al., 2011) and has the advantage of generating metal byproducts for later use. However, transporting sufficient masses of hydrogen to the Moon (about 100 kg of hydrogen for every metric ton of concrete produced) could be cost prohibitive.

In the absence of water, another binder already available on the Moon must be identified for concreting to be feasible. One candidate could be sulfur, which is present in troilite (FeS, about 1% of lunar regolith by mass). If it can be extracted, sulfur has a low melting point (about 115 °C at 1 bar pressure) and could bind aggregate media if mixed as a liquid and re-solidified. In fact, sulfur binders have been investigated in exploratory studies (I. Casanova 1997). However, molten sulfur is highly volatile, noxious, and even toxic. In addition, solidified sulfur mass is quite soft (Mohs hardness of 2) and has poor durability under thermal cycling.

Sintering Methods

Sintering takes advantage of the embodied surface energy in a fine powder mass, providing thermal energy to give the atoms enough mobility, at about 60% to 80% of the melting temperature, to rearrange in configurations with lower surface energy. The result is a fusing of the grains together into a rigid network with reduced porosity.

Most of the mineral constituents of lunar regolith are aluminosilicates with relatively low melting temperatures (800 °C to 1000 °C) compared to many technical ceramics on Earth. This, combined with the primarily ionic character of their chemical bonds, makes them relatively easy to sinter.

Besides this, sintering avoids the two greatest challenges of concrete because it requires no water and can be easily performed in the vacuum of space. In addition, the high-volume fraction of glassy pyroclastic and agglutinated material in lunar regolith likely become viscous liquids, which can provide an additional driving force for reducing porosity through capillary attraction. Even so, at least three potential technical barriers must be assessed before sintering can become widely used on the Moon. First, unlike concrete, the powder must be quite fine to impart enough surface energy. Lunar regolith has an average particle size of about 70 μm, whereas most "sinterable" ceramic and metal powders are at least 10 times finer. Some computer simulations, combined with limited experimentation, will need to be done to determine how much densification is possible with such coarse particles. Greater temperatures, perhaps even some partial melting, may be required to obtain enough mechanical integrity in sintered parts for lunar

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applications. Recently, overcoming the challenge has been explored by using the spark plasma sintering (SPS) method. Unlike conventional sintering, the SPS employs a pulse electric current, an applied uniaxial pressure, and rapid heating rate in a vacuum atmosphere to sinter ceramic or metal powders. It could provide many advantages due to the combined effect of electric field and applied pressure, such as high heating and cooling rates, short processing times, low sintering temperatures, and a high density of sintered products (nearly 100% theoretical density). The SPS method has been examined for its feasibility as a potential ISRU construction technology, and results using a lunar simulant FJS-1 demonstrate that it is a quite promising method to effectively densify lunar soils for structural applications in space exploration (Zhang et al. 2019). Second, powders with high vapor pressures below the melting point are often difficult to sinter because they tend rather to coarsen by evaporation and condensation rather than densify into a low-porosity solid. And finally, sintering of large elements can be limited by the size of the furnace used or, in the case of using focused solar radiation or lasers, by the rate of heat transfer to the interior of the mass. Again, thermodynamic modeling and finite element computer simulations can be used to forecast the extent to which these issues are likely to impede the production rate or size of construction elements by sintering methods.

METHODOLOGY

The research team is currently developing and validating microstructure models for concrete materials, and soon will begin adapting microstructure models for sintering of lunar regolith components. The model for concrete, called Thermodynamic Hydration and Microstructure Evolution Simulator (THAMES), relies heavily on geochemical thermodynamic methods for predicting the stable mineralogical components in a mixture under prescribed environmental conditions of temperature and pressure. The thermodynamic predictions of mineral volume fractions are then used to inform 3D microstructure algorithms that arrange the components in space to achieve realistic "virtual" microstructures times (Bullard and Garboczi, 2009). It leverages thermodynamic predictions of equilibrium phase assemblages in aqueous geochemical systems to construct 3-D virtual microstructures of a cementitious binder at different times during the hydration process or during degradation phenomena. These virtual microstructures can then be used directly to calculate important engineering properties of a concrete made from that binder at prescribed mixture proportions and environmental conditions. The microstructure aspect of the model is important because the mechanical properties of the material can be inferred from direct application of finite element methods on the microstructures. Of equal importance, this method will assess the potential for development of any deleterious product development that would hamper the success of the product.

THAMES has previously been used for various applications of Portland cement concrete, from computing heats of hydration and compressive strength (Bullard et al., 2011) to drying shrinkage (Li et al., 2017) and viscoelastic properties (Li et al., 2018). Most lunar regolith mineral components, or close analogs, have already been integrated in thermodynamic databases (MINES Thermodynamic Database available at http://tdb.mines.edu/). These data should be useful for application to lunar concrete, whether by water-based or sulfur-based binders. However, the modeling of these components will require knowledge of any kinetic constraints that prevent the establishment of true equilibrium. This knowledge can only be acquired by experimental characterization of the materials during the process by techniques such as time-dependent quantitative X-ray diffraction. Moreover, additional characterization will need to be

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made of the glassy, agglutinated components in regolith to determine their dissolution rates in the relevant liquid environment. Finally, more experiments will need to be conducted to gain insights on microstructure evolution in these systems so that the algorithms in THAMES can be modified if necessary to produce high-fidelity microstructures. However, all such experiments only require small quantities of homogenized material, on the order of grams.

Materials and processing characterization

Agglutinates

The properties of lunar regolith afford new opportunities and challenges for in situ construction that highlight the importance of using physically and chemically accurate simulants in materials testing. Current efforts to return to the Moon require new geomaterial characterization, analysis, and fabrication for construction, engineering, and resource extraction purposes in the unique environment on the surface of the Moon. This need has led to the proliferation of simulants, which are synthesized terrestrial materials, designed to mimic the chemical, physical, and mechanical properties of the lunar soil. JSC- 1A is the simulant used in most studies, because it is the most readily available, but JSC-1A is more mineralogically similar to the mare regions. Upcoming lunar return missions are targeting the lunar highland regions however, so more studies need to be conducted using highlands simulants which they have distinct physical and mineralogical properties. Due to higher rates of meteoritic bombardment in the highlands, there is also a higher percentage composition of agglutinates in the highlands simulants. Very few, if any, of the studies conducted have used agglutinated simulants however, because of manufacturing challenges.

Two companies— Off Planet and Outward Technologies (formerly Blueshift)— now make lunar simulants with agglutinates, and work is needed for third-party full geomaterial characterization of these simulants. This method would examine chemical, physical, and geotechnical properties of the agglutinate simulants using a range of laboratory techniques including x-ray diffraction, x-ray fluorescence, Raman spectroscopy, scanning electron microscopy, petrographic microscopy, particle size and shape analysis as well as conventional geotechnical laboratory tests to determine compaction characteristics and shear strength parameters. Using this information, polymerization/modification of the simulants will be done to develop materials for construction specific applications such as space craft landing pads, roving vehicles, and 3D-printing mixtures. Figure 3a and 3b show microscopic images of lunar agglutinate simulants with realistic grain shape consisting of highly irregular features. The agglutinate grain sizes that have been observed range from 200 µm to centimeter scale, though there are likely grains as small as 40 µm, again underscoring the need for characterization.

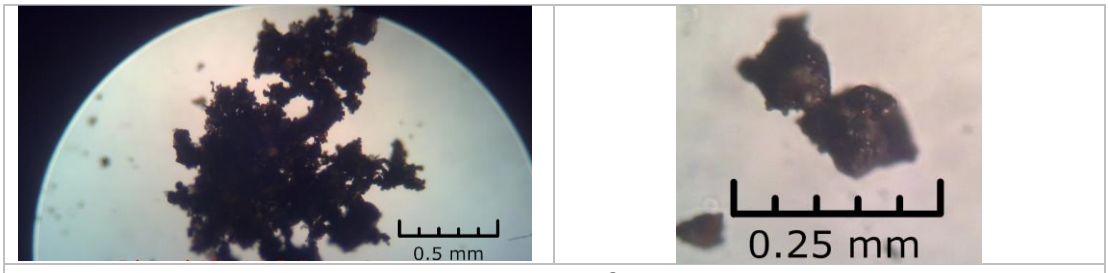

Fig3a and 3b. Microscopic images of lunar agglutinate simulants (Used with permission: Outward Technologies)

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Basalt Glass Fiber

Basalt glass fiber can be created on the Moon by melting and drawing the lunar regolith into fiber. This could be an incredibly valuable material for construction and manufacturing on the Moon. Polymer binders would be made available in the form of astronaut trash, or bioplastics created from crops and algae that will be necessary for a human presence on the Moon. It is also possible to eliminate the polymer binders from the additive construction process completely. By directly sintering granular material or drawn glass fiber, structures can be created in situ without the need for polymers (Buckles et al., 2019). It is proposed that studies be conducted to understand processing considerations for making basalt glass fibers on the Moon. In particular, it is important to understand what unusual characteristics may result from the 1/6 gravity lunar environment.

Document challenges of the Earth-based laboratory

Changes in soil from Moon to laboratory should be considered along with some critical differences between simulants and lunar soil. The high content of water vapor and oxygen in the laboratory causes more water to be adsorbed on grain surfaces and results in oxidation. The abundant metallic iron in lunar soils rusts easily when oxidized. (Allton et al., 1985). Discrete element modeling can be used to study materials properties in 1/6 gravity which could be further informed by parabolic or sub-orbital testing.

CONCLUSION

Working with researchers using materials science applications for improved infrastructure materials, this research will substitute accurate regolith simulant for terrestrial materials in preparation for using this amalgamated regolith material for construction experiments for future lunar base construction. Developing hybrid synthesized materials for future lunar construction will be evaluated for best practices and according to a variety of requirements and criteria such as energy intensity, strength factors, time to construct, perceived cost, and others. The objective of this research is to begin to identify the extent to which existing models and methods for developing and testing new geomaterials can be adapted to evaluate regolith-based novel materials under lunar environmental conditions. The ultimate intent is to develop an ASTM or ASCE approved standard for future ISRU activities through the means of analysis in the preliminary stage and then tests to achieve the standard process. This work will further ISRU for lunar construction in line with NASA's recent call to action to return to the Moon "to stay" by 2024.

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